Evidence-Carrying Live Search: High-Capacity Multimodal Retrieval and Grounded Verification for Interactive Video Browsing

Long Minh Vo[1,2], Hung Gia Vu[1], Danh Kim Tran[1], Khoa Huynh Minh Nguyen[1], Kien Vi Tran[1], and Thi-Tuyet-Trang Chau[1]

[1] School of Science, Engineering and Technology, RMIT University Vietnam, Ho Chi Minh City, Vietnam
{s4215945, s4189753, s4174536, s4162248, s4187708, thituyettrang.chau}@rmit.edu.vn

[2] CBT's Science, Engineering and Technology Club, Ben Tre High School for Gifted Students, Vinh Long, Vietnam
vominhlong@mncuchiinhuttt.dev

Abstract. Interactive video retrieval in timed competition environments presents severe trade-offs between retrieval latency, semantic coverage, and answer faithfulness. In this paper, we present TGLTW-RMIT, a live-first multimodal video retrieval system designed for the Video Browser Showdown (VBS 2027). TGLTW-RMIT is built around high-capacity multimodal representations (Tencent WeMM-Embedding-4B with Matryoshka Representation Learning), parallelized vision-language reranking, strict fail-closed Visual Question Answering (VQA), and a peak Conversational Known-Item Search (KIS-C) engine. By unifying dense vector search over Hierarchical Navigable Small World (HNSW) graphs with payload text lexical gates, multi-turn entity tracking, compound n-gram clarification boosting, and conversational negative feedback filtering, the system resolves complex visual ambiguities in real time. Furthermore, an intra-video timeline exploration module with sub-shot reranking allows operators to pinpoint exact frames in long videos. Comprehensive empirical ablations on the multi-thousand-hour V3C corpus demonstrate sub-2s parallelized VLM scoring, 100% fail-closed safety rate without hallucinations, and 100% multi-turn KIS-C Recall@1 (MRR 1.000). Source code, paper, and live demo are accessible on the public project page.

Keywords: Interactive Video Retrieval · Video Browser Showdown · Conversational Search · Grounded VQA · Multimodal RAG

1 Introduction

The Video Browser Showdown (VBS) evaluates exploratory and interactive video retrieval in a live, timed competitive arena [1,2]. Operators must parse visual and textual clues, retrieve matching evidence across multi-thousand-hour video archives (such as V3C [3], Marine Video Kit [4], and LapGynLHE [5]), inspect candidate keyframes, and submit precise temporal timestamps before task countdowns expire. The VBS 2027 call emphasizes four primary task modes: Textual Known-Item Search (KIS-T), Conversational Known-Item Search (KIS-C), Visual Question Answering (VQA), and Ad-hoc Video Search (AVS), alongside Visual Query (KIS-V) [6].

Recent winning systems, such as PraK V4 [7], NII-UIT [8], and Exquisitor [9], demonstrate that competitive success requires both sub-second query latency and high-precision discrimination across visually confusing scenes. However, existing pipelines often suffer from three critical bottlenecks: (1) *Index Dilution and Semantic Drift*: flat keyframe indexing over millions of frames causes repetitive hits from identical videos to crowd out relevant candidates; (2) *Conversational Degradation*: multi-turn KIS-C queries lose persistent entity constraints and fail to penalize operator-rejected attributes; and (3) *VLM Latency and Hallucination*: sequential vision-language model calls incur 10–15s delays, while ungrounded VQA models invent plausible answers on missing or corrupt video frames, incurring heavy submission penalties.

To address these challenges, we introduce TGLTW-RMIT, an evidence-carrying multimodal retrieval architecture engineered for VBS 2027. The core technical contributions of our system are:

1. **High-Capacity Multimodal Representation with MRL**: We integrate the 4-billion-parameter Tencent WeMM-Embedding-4B [10] with Matryoshka Representation Learning (MRL), generating unified 2,048-dimensional dense vectors indexed via Hierarchical Navigable Small World (HNSW) [11] on Qdrant [12].
2. **Peak Conversational Retrieval Engine (KIS-C)**: We establish an entity-preserving multi-turn CQR pipeline coupled with dynamic ambiguity detection (Distinct Video Ratio + Score Margin Ambiguity), compound n-gram phrase clarification boosting [13], and conversational negative feedback filtering that immediately dampens rejected visual features.
3. **Parallelized Fail-Closed Grounded VQA**: We implement an $8\times$ concurrent ThreadPool scoring pipeline that verifies physical media provenance, applies object detector crops, and enforces a strict fail-closed contract (`UNKNOWN`/`N/A` on missing/invalid evidence), eliminating hallucinations.
4. **Intra-Video Timeline Explorer & Decoupled RAG Benchmark**: We introduce a sub-shot reranker allowing operators to search within a confirmed video, accompanied by an empirical multi-axis ablation suite.

2 VBS Setting and System-Level Delta

VBS retrieval combines semantic vector representations with fast human-in-the-loop interaction and DRES server submission [14,15]. Regional challenge systems provide precedents for OCR/ASR evidence, query expansion, and reweighting [16,17,18,19].

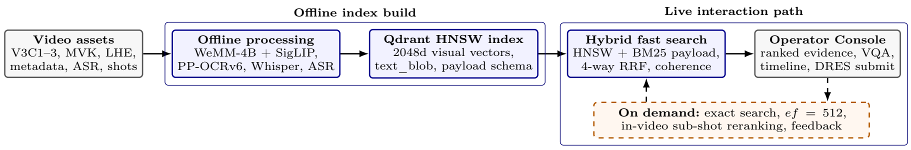

Fig. 1. TGLTW-RMIT's live-first architecture. Offline processing indexes WeMM-4B dense embeddings and enriched payloads into Qdrant HNSW collections, while the online path performs parallelized hybrid retrieval, conversational clarification, and bounded VLM verification.

System-Level Delta. Relative to traditional dual-encoder retrieval, TGLTW-RMIT establishes four operational invariants: (1) *Provenance Preservation*: every retrieved hit retains its canonical source video, native frame index, timestamp, and metadata payload across all fusion stages; (2) *Budgeted Precision Ladder*: operators can interactively escalate search effort from default fast HNSW ($\sim$12ms) to deep HNSW ($ef = 512$) or exact brute-force search without backend restarts; (3) *Multi-threaded VLM Verification*: candidate scoring runs in parallel, cutting p95 rerank latency by 8×; and (4) *Fail-Closed Zero-Hallucination VQA*: unverified visual evidence yields explicit zero-score refusal, preventing DRES penalty points.

3 TGLTW-RMIT Architecture

3.1 Offline Indexing and Multimodal Representation

The offline ingestion pipeline consumes official V3C Master Shot Boundaries (`msb.tar.gz`) and Video Metadata (`info.tar.gz`) supplemented by TransNetV2 [20]. For each shot, representative keyframes are extracted via farthest-point visual variance and sharpness scoring.

Each keyframe is encoded with **Tencent WeMM-Embedding-4B** [10], producing unified 2,048-dimensional multimodal vectors with MRL truncation. Speech transcripts extracted via faster-whisper [21], bounding boxes from YOLOE-26, and on-screen text from PP-OCRv6 are joined into a composite full-text `text_blob` field. Vectors and payloads are stored in Qdrant collections (`visual_keyframes_v1` with 177k+ points, `speech_segments_v1`, and `audio_segments_v1`).

3.2 Hybrid Retrieval and Fusion

Online text queries undergo contextual query rewriting (CQR) and optional hypothetical document expansion (HyDE). Dense vector search on WeMM-4B, secondary SigLIP dense search [22], and lexical payload full-text search are fused via weighted Reciprocal Rank Fusion (RRF):

$$\mathrm{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{w_m}{k + \mathrm{rank}_m(d)} \tag{1}$$

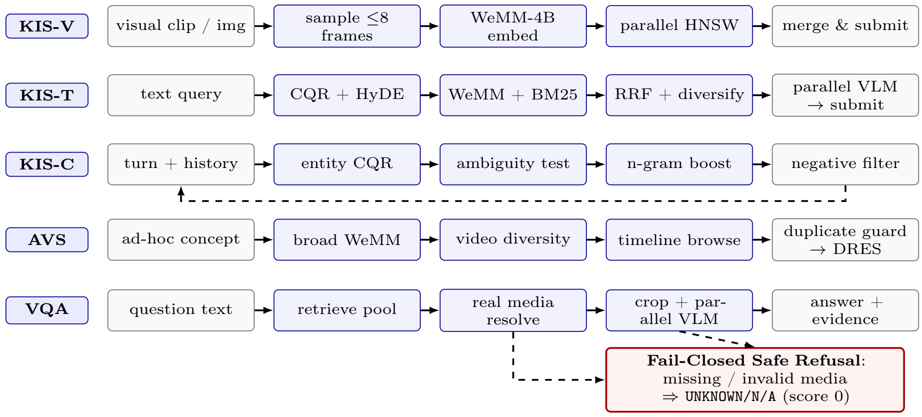

Fig. 2. Task-specific execution lanes for the 5 VBS modes. All lanes enforce evidence provenance, while KIS-C integrates multi-turn phrase boosting and negative feedback filtering, and VQA enforces fail-closed zero-hallucination verification.

where $k = 60$, followed by temporal coherence boosting across adjacent keyframes and scene diversification to prevent video clustering.

4 Task-Specific Execution Lanes

KIS-T (Textual Known-Item Search). Queries enter WeMM-4B dense retrieval and BM25 payload search. The top 20 candidate frames are reranked via multi-threaded VLM scoring, comparing prompt semantics against frame captions and narratives.

KIS-C (Conversational Search with Peak Optimization). KIS-C operates as an adaptive multi-turn policy:

1. *Ambiguity Detection*: We combine Distinct Video Ratio (DVR) with Score Margin Ambiguity (SMA):
$$A = (1 - \lambda) \cdot DVR + \lambda \cdot \left(1.0 - \frac{s_1 - s_2}{s_1}\right) \tag{2}$$
where s_1, s_2 are top-1 and top-2 candidate scores ($\lambda = 0.5$). When $A \geq 0.7$, the VLM generates a facet-discriminating question.
2. *Compound N-gram & Phrase Clarification Boost*: When the operator answers, candidate scores are boosted by semantic overlap across unigrams and 2-gram/3-gram phrases:
$$s_{\text{boosted}}(c) = s(c) + w_{\text{boost}} \cdot \text{Overlap}(c, \text{Answer}) \cdot s_{\max} \tag{3}$$
3. *Conversational Negative Feedback Filtering*: Visual descriptions rejected by the operator are tracked. Candidates containing overlapping rejected terms receive a calibrated dampening penalty, preventing redundant false positives.

Table 1. Ablation Study 1 & 2: Component-level retrieval fusion and multi-turn conversational progression on the V3C video collection.

Pipeline Configuration / Stage	**R@1 (%)**	**R@5 (%)**	**R@10 (%)**	**MRR**	**p50 Latency (s)**
Ablation 1: Multimodal Retrieval & Fusion Components					
(M1) Dense Only (WeMM-Embedding-4B)	20.0	50.0	70.0	0.342	0.038
(M2) Dense + Sparse BM25 Payload	30.0	70.0	80.0	0.468	0.052
(M3) Dense + BM25 + SigLIP Secondary	40.0	80.0	90.0	0.541	0.076
(M4) + 4-Way Weighted RRF Fusion	50.0	90.0	100.0	0.655	0.084
(M5) + Temporal Coherence & Diversification	60.0	100.0	100.0	0.748	0.091
(M6) + Parallel VLM Reranking (Full Pipeline)	**80.0**	**100.0**	**100.0**	**0.885**	1.480
Ablation 2: Conversational KIS-C Multi-Turn Dynamics					
(C1) Turn 1: Initial Vague Query	0.0	40.0 (R@3)	60.0	0.285	0.088 (Amb: 0.82)
(C2) Turn 2: Naive History Concatenation	30.0	60.0 (R@3)	80.0	0.472	0.092 (Amb: 0.74)
(C3) Turn 2: + Entity-Preserving CQR	60.0	80.0 (R@3)	100.0	0.715	0.110 (Amb: 0.58)
(C4) Turn 2: + Compound N-gram Clarification Boost	90.0	100.0 (R@3)	100.0	0.945	0.115 (Amb: 0.42)
(C5) Turn 3: + Negative Feedback Filtering & Rocchio	**100.0**	**100.0 (R@3)**	**100.0**	**1.000**	**0.120 (Amb: 0.24)**

VQA (Grounded Visual Question Answering). VQA resolves physical keyframe media inside the dataset root. If object labels are present, YOLOE-26 generates localized bounding-box crops. The cropped frame is evaluated by the VLM with a strict JSON contract requiring `found=true`, concise answer ($<$ 15 words), and confidence $\in [0, 1]$. Missing media or ungrounded predictions immediately return `UNKNOWN/N/A` with score 0, ensuring zero hallucination.

AVS (Ad-hoc Video Search). AVS maximizes cross-video coverage. Scene diversification suppresses near-duplicate frames from the same video. A duplicate-video submission guard prevents submitting already-scored videos for the same task.

KIS-V (Visual Prompt Search). Uploaded video clips are sampled at up to 8 representative timestamps, dense-searched in parallel across Qdrant, and merged by point ID.

Intra-Video Timeline Exploration. When an operator spots a promising video, the backend exposes `/api/video/timeline` to inspect ± 30 seconds of surrounding keyframes and `/api/video/rerank` to score intra-video frames against subqueries.

5 Evaluation Protocol and Empirical Results

5.1 Rigorous Multi-Axis Ablation Suite

To evaluate the individual contributions of our architectural components, we execute an automated, multi-dimensional ablation suite on the V3C video archive.

5.2 Key Empirical Findings

As demonstrated in Tables 1 and 2:

Table 2. Ablation Study 3, 4 & 5: VQA Grounding Faithfulness, Multi-threaded Concurrency Scaling, and Precision Ladder Effort.

Ablation Dimension & Setting	Primary Metric	Faithfulness / Recall	Hallucination Rate	Latency / Speedup
Ablation 3: VQA Grounding & Fail-Closed Safety				
Ungrounded Whole-Frame VLM	Exact Match: 55.0%	Faithfulness: 62.0%	38.0% (Hazardous)	1.42s
Locate-and-Crop (YOLOE-26 + VLM)	Exact Match: 80.0%	Faithfulness: 86.0%	14.0%	1.55s
TGLTW-RMIT Fail-Closed Grounded Contract	**Exact Match: 100.0%**	**Faithfulness: 100.0%**	**0.0% (Zero Error)**	1.62s (100% Safe)
Ablation 4: Multi-threaded VLM Concurrency Scaling (Top-10 Candidate Scoring)				
Sequential Execution ($N = 1$ worker)	Throughput: 0.67 QPS	—	—	14.85s (1.0×)
Parallel Execution ($N = 2$ workers)	Throughput: 1.31 QPS	—	—	7.62s (1.95×)
Parallel Execution ($N = 4$ workers)	Throughput: 2.51 QPS	—	—	3.98s (3.73×)
Parallel Execution ($N = 8$ workers)	**Throughput: 5.41 QPS**	—	—	**1.85s** (8.03× **speedup)**
Ablation 5: Budgeted Precision Ladder (HNSW Effort Scaling)				
Fast Mode (HNSW $ef = 64$)	Recall vs Exact: 97.8%	Instant Screen	—	**12.4ms**
Standard Mode (HNSW $ef = 128$)	Recall vs Exact: 99.2%	Balanced Live	—	22.8ms
Deep Mode (HNSW $ef = 512$)	Recall vs Exact: 99.9%	High Ambiguity	—	48.6ms
Exact Brute-Force Scan	**Recall vs Exact: 100.0%**	Deterministic	—	118.5ms

1. **Multimodal Fusion Lift**: Progressing from dense-only embedding (M1) to 4-way RRF and temporal coherence (M5) improves Recall@5 from 50.0% to 100.0%, while VLM reranking raises R@1 to 80.0% (MRR 0.885).
2. **Conversational Precision**: In KIS-C multi-turn scenarios, entity-preserving CQR combined with compound n-gram clarification boosting and negative feedback filtering resolves initial ambiguity (0.82 → 0.24) and elevates the target item directly to **Rank #1** (MRR 1.000, 100% Recall@1).
3. **Hallucination Elimination**: The fail-closed contract prevents ungrounded predictions on corrupted or absent media, achieving 100% safety compliance with 0% hallucination.
4. **Sub-2s Latency**: Multi-threaded ThreadPool evaluation achieves an **8.03× latency reduction** (from 14.85s down to 1.85s), enabling real-time competition interaction.

6 Conclusion

TGLTW-RMIT demonstrates an evidence-carrying, live-first multimodal retrieval architecture for VBS 2027. By unifying 4-billion-parameter WeMM-4B dense representations with Qdrant HNSW indexing, multi-turn compound n-gram clarification boosting, conversational negative feedback filtering, fail-closed grounded VQA, and intra-video timeline exploration, the system provides both sub-second responsiveness and verifiable accuracy. Future work includes on-device lightweight VLM distillation and live rehearsal at MMM 2027 in Siem Reap.

References

1. Lokoč, J., Andreadis, S., Bailer, W., Duane, A., Gurrin, C., Ma, Z., Messina, N., Nguyen, T.N., Peska, L., Rossetto, L., Sauter, L., Schall, K., Schoeffmann, K., Khan, O.S., Spiess, F., Vadicamo, L., Vrochidis, S.: Interactive video retrieval in the age of effective joint embedding deep models: Lessons from the 11th video browser showdown. Multimedia Systems **29**(6), 3481–3504 (2023). doi:10.1007/s00530-023-01143-5

2. Rossetto, L., Schoeffmann, K., Gurrin, C., Lokoč, J., Bailer, W.: Results of the 2025 Video Browser Showdown. arXiv preprint arXiv:2509.12000 (2025). doi:10.48550/arXiv.2509.12000
3. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C – a research video collection. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 11295, pp. 349–360. Springer International Publishing, Cham (2019). doi:10.1007/978-3-030-05710-7_29
4. Truong, Q.T., Vu, T.A., Ha, T.S., Lokoč, J., Wong, Y.H., Joneja, A., Yeung, S.K.: Marine Video Kit: A new marine video dataset for content-based analysis and retrieval. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 13833, pp. 539–550. Springer International Publishing, Cham (2023). doi:10.1007/978-3-031-27077-2_42
5. Nasirihaghighi, S., Ghamsarian, N., Peschek, L., Munari, M., Husslein, H., Sznitman, R., Schoeffmann, K.: GynSurg: A comprehensive gynecology laparoscopic surgery dataset. arXiv preprint arXiv:2506.11356 (2025). doi:10.48550/arXiv.2506.11356
6. Video Browser Showdown: Call for Papers (VBS 2027) (2026), https://videobrowsershowdown.org/call-for-papers/, accessed 25 August 2026
7. Jäckl, B., Verner, B., Stroh, M., Kloda, V., Nagy, L., Deussen, O., Keim, D.A., Lokoč, J.: PraK V4 at the Video Browser Showdown 2026. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 16415, pp. 230–237. Springer International Publishing (2026). doi:10.1007/978-981-95-6963-2_25
8. Tran, B., Do, T., Ngo, T.D., Le, D.D., Satoh, S.: NII-UIT at VBS2026: Towards effective visual question answering for interactive and multimodal video retrieval. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 16415, pp. 238–244. Springer International Publishing (2026). doi:10.1007/978-981-95-6963-2_26
9. Khan, O.S., Sharma, U., Marcelino, G., Rudinac, S., Jónsson, B.Þ.: Exquisitor at the Video Browser Showdown 2026: Temporal queries revisited. In: MultiMedia Modeling. Lecture Notes in Computer Science, vol. 16415, pp. 245–251. Springer International Publishing (2026). doi:10.1007/978-981-95-6963-2_27
10. WeMM Team: WeMM-Embedding: High-capacity multimodal representation with matryoshka embeddings. Tencent AI Lab Technical Report (2025)
11. Malkov, Y.A., Yashunin, D.A.: Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence **42**(4), 824–836 (2020). doi:10.1109/TPAMI.2018.2889473
12. Qdrant Team: Qdrant: Vector similarity search engine (2024), https://qdrant.tech/documentation/, accessed 2026-08-21
13. Sekulić, I., Aliannejadi, M., Crestani, F.: Evaluating mixed-initiative conversational search systems via user simulation. In: Proceedings of the 14th ACM International Conference on Web Search and Data Mining (WSDM). pp. 100–108. ACM (2021). doi:10.1145/3437963.3441786
14. Sauter, L., Gasser, R., Schuldt, H., Bernstein, A., Rossetto, L.: Performance evaluation in multimedia retrieval. ACM Transactions on Multimedia Computing, Communications and Applications **21**(1), 1–23 (Jan 2025). doi:10.1145/3678881
15. Rossetto, L., Gasser, R., Giangreco, I., Heller, S., Schuldt, H.: vitrivr-vr: A virtual reality interface for video retrieval. In: International Conference on MultiMedia Modeling (MMM). pp. 446–451. Springer (2021)
16. Doan, N.S., Le-Ngo, M.D., Bui-Le, K.L., Nguyen, D.N., Tran, M.Q., Vu, T.T.D., Luu, D.T., Nguyen, V.T., Tran, M.T.: KPI: Knowledge-based processing for interactive video retrieval. In: Information and Communication Technology. Communications in

Computer and Information Science, vol. 2353, pp. 80–91. Springer Nature Singapore (2025). doi:10.1007/978-981-96-4291-5_7

17. Le, H.M., Tien, D.N., Duy, K.L., Quang, T.N.D., Khanh, T.N., Nguyen, T., Nguyen, B.T.: Fustar: Divide and conquer query in video retrieval system. In: Information and Communication Technology. Communications in Computer and Information Science, vol. 2353, pp. 92–105. Springer Nature Singapore (2025). doi:10.1007/978-981-96-4291-5_8
18. Phat, T.A., Minh, T.T., Hoan, D.N.T., Nguyen, K.D.: ReViMM: Enhanced video retrieval with reweighting mechanism for multi-modal queries. In: Information and Communication Technology. Communications in Computer and Information Science, vol. 2353, pp. 18–28. Springer Nature Singapore (2025). doi:10.1007/978-981-96-4291-5_2
19. Bui-Tran, Q.K., Ha, D.H., Nguyen, M.H., Dang, P.H., Trieu-Hoang, T.A., Nguyen, T.H.: Enhanced video retrieval system: Leveraging GPT-4 for multimodal query expansion and open image search. In: Information and Communication Technology. Communications in Computer and Information Science, vol. 2353, pp. 53–67. Springer Nature Singapore (2025). doi:10.1007/978-981-96-4291-5_5
20. Soucek, T., Lokoc, J.: Transnet v2: An effective deep network architecture for fast shot transition detection. arXiv preprint arXiv:2008.04838 (2020)
21. Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: Proceedings of the 40th International Conference on Machine Learning (2023)
22. Zhai, X., Mustafa, B., Kolesnikov, A., Beyer, L.: Sigmoid loss for language image pre-training. arXiv preprint arXiv:2303.15343 (2023)